

Exploring the Seven Domains of a Typical IT Infrastructure (4e)

Fundamentals of Information Systems Security, Fourth Edition - Lab 01

Student:

Hassinullah Niazy

Email:

niazy27@up.edu

Time on Task:

2 hours, 58 minutes

Progress:

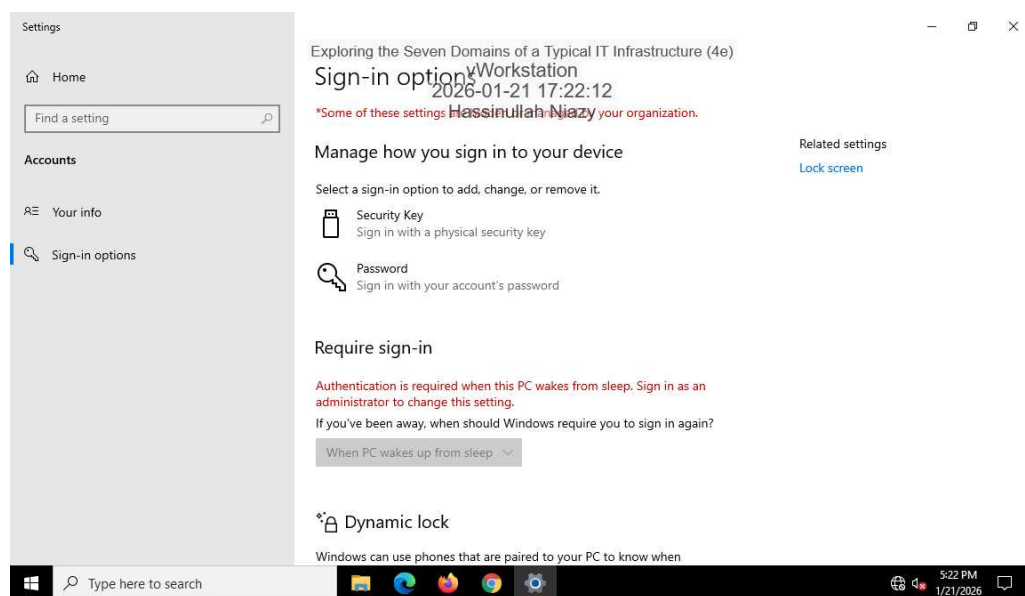
100%

Report Generated: Friday, January 23, 2026 at 2:14 AM

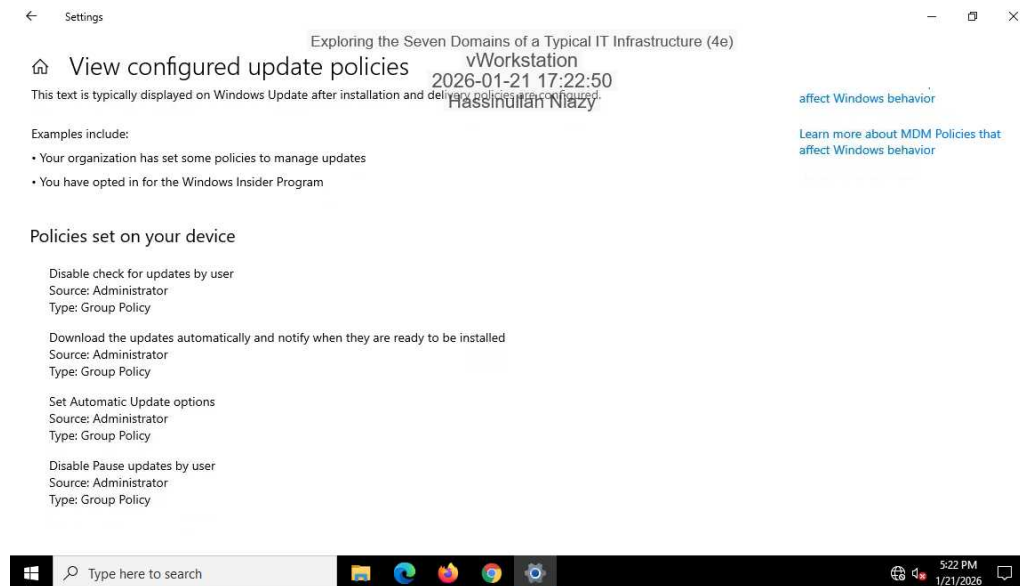
Section 1: Hands-On Demonstration

Part 1: Explore the Workstation Domain

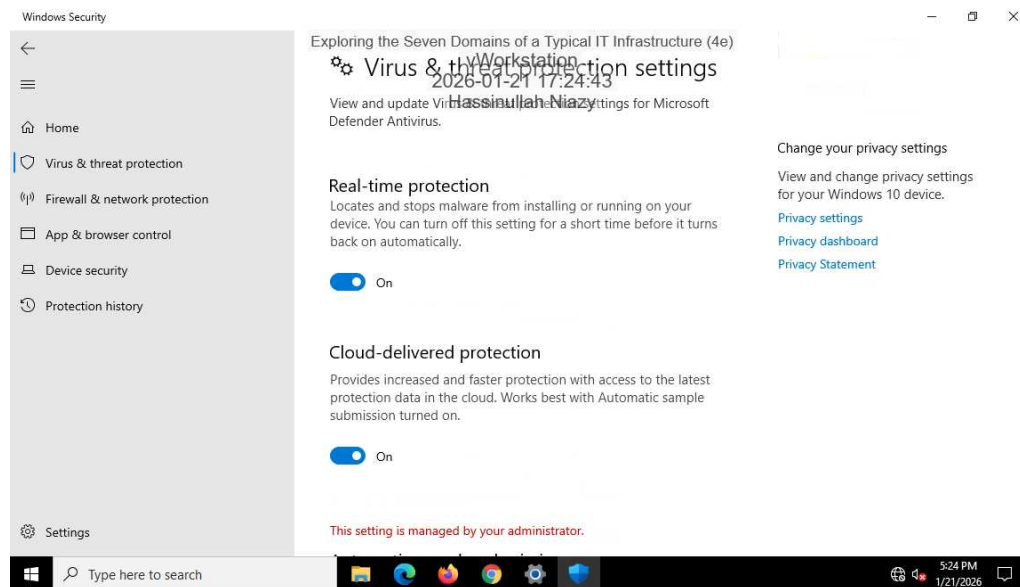
4. Make screen capture showing the **Sign-in options** for Alice's account.



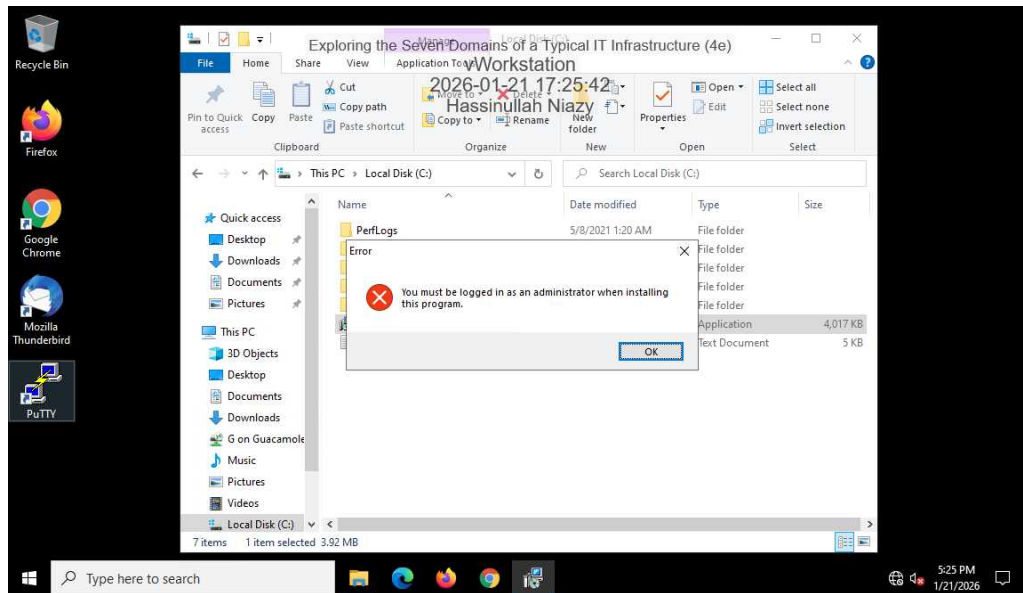
7. Make a screen capture showing the View configured update policies page.



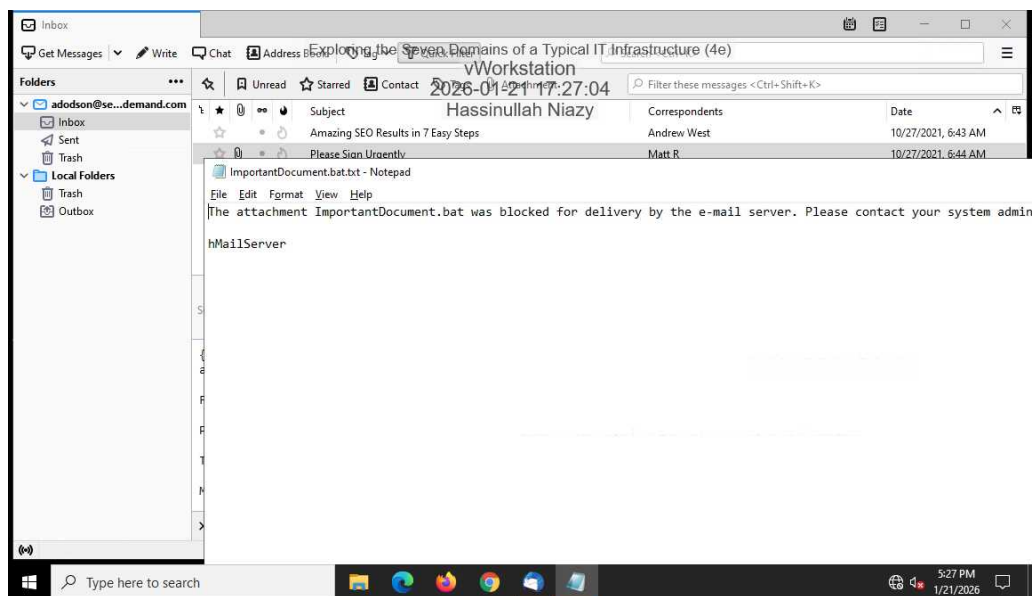
14. Make a screen capture showing the Virus & Threat Protection Settings.



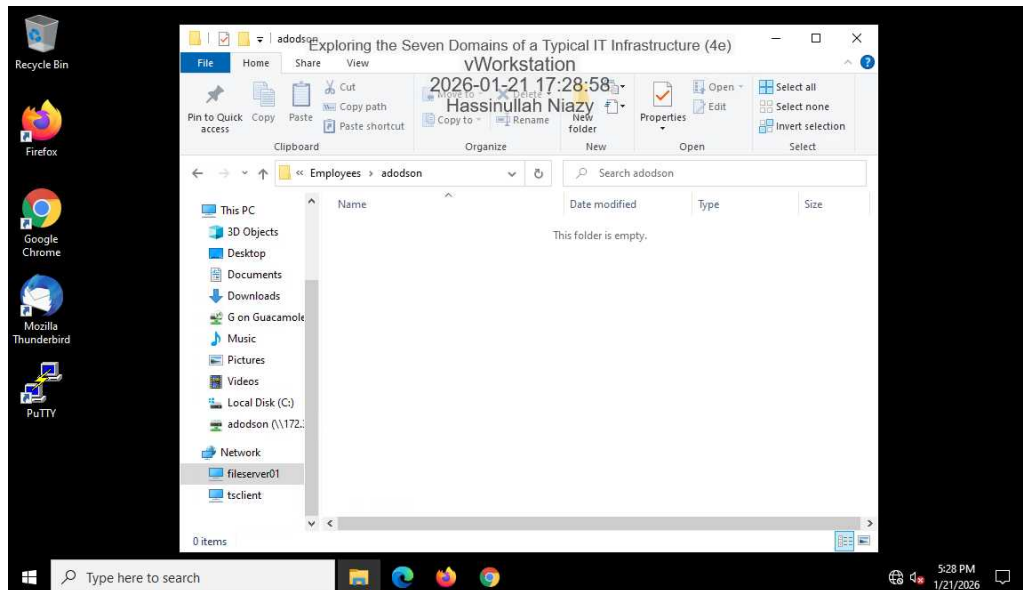
18. Make a screen capture showing the security warning from attempting to run an executable file.



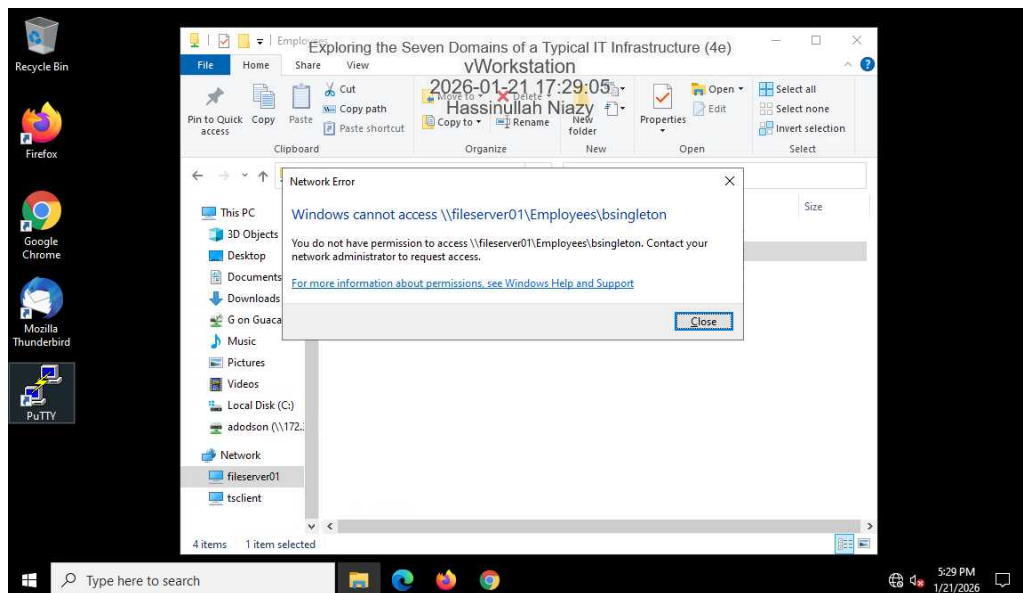
24. Make a screen capture showing the blocked attachment message.



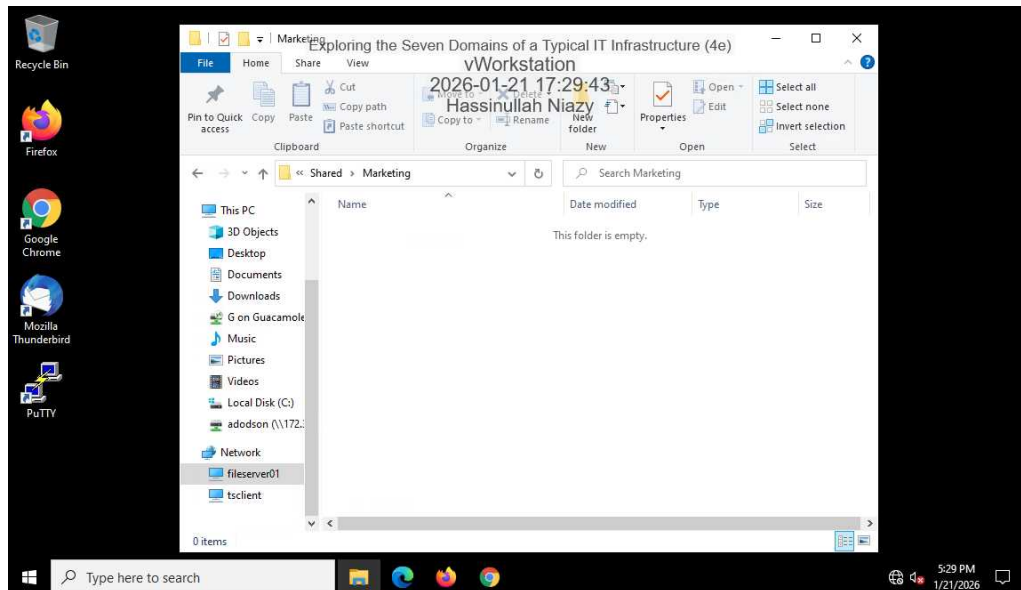
28. Make a screen capture showing a successful connection to the adodson user folder.



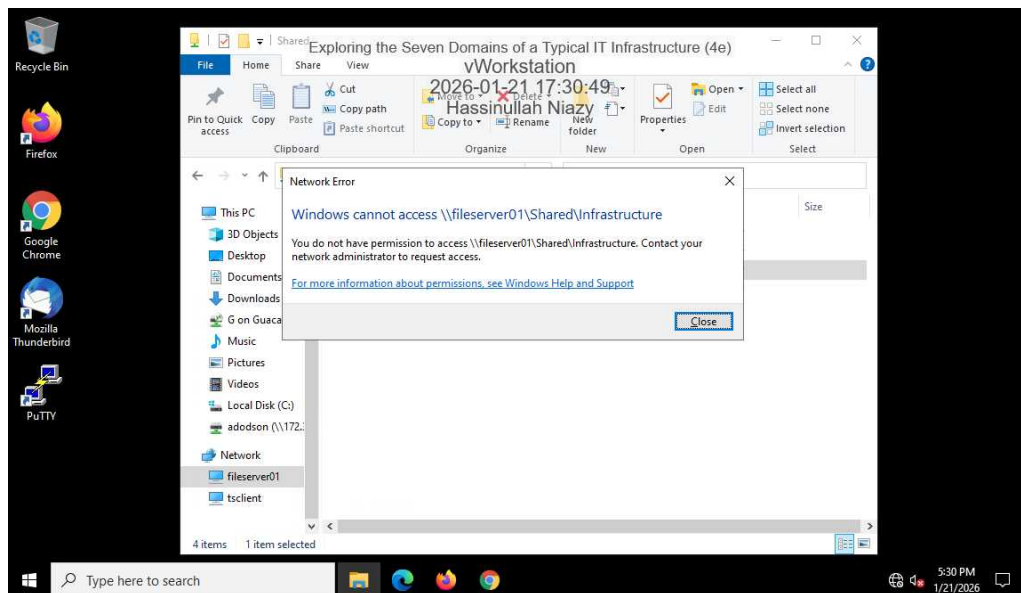
29. Make a screen capture showing a failed connection to another user folder.



31. Make a screen capture showing a successful connection to the Marketing shared folder.

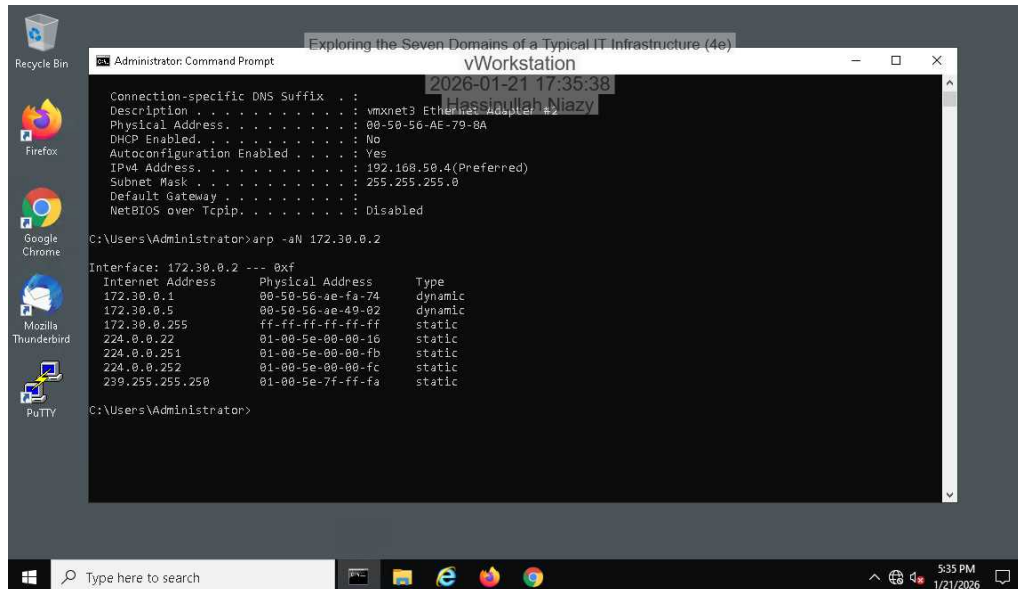


32. Make a screen capture showing a failed connection to another shared folder.

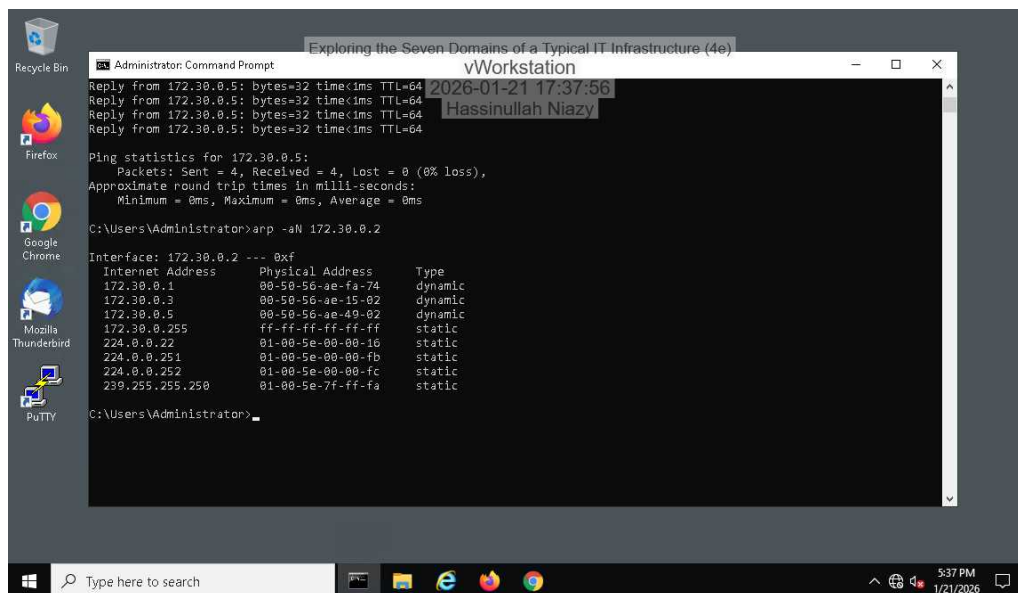


Part 2: Explore the LAN Domain

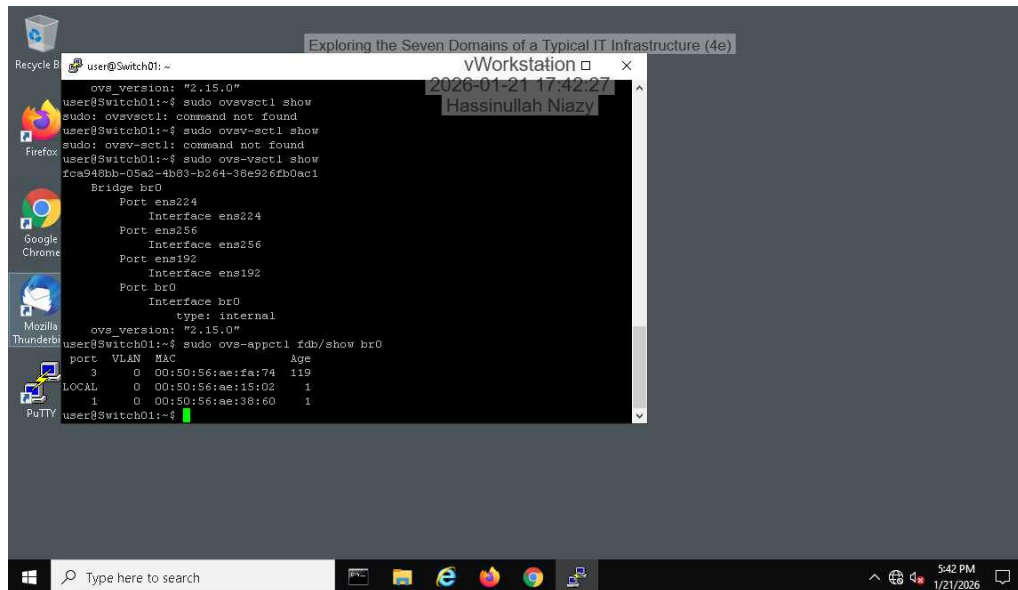
5. Make a screen capture showing the vWorkstation's original ARP table.



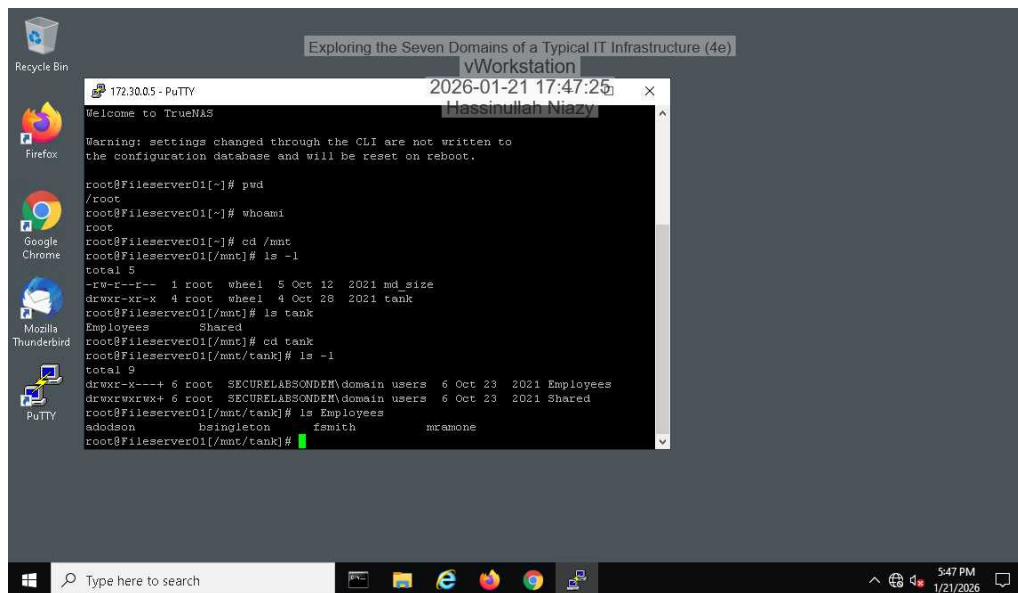
10. Make a screen capture showing the vWorkstation's updated ARP table.



20. Make a screen capture showing the Switch01 forwarding table.

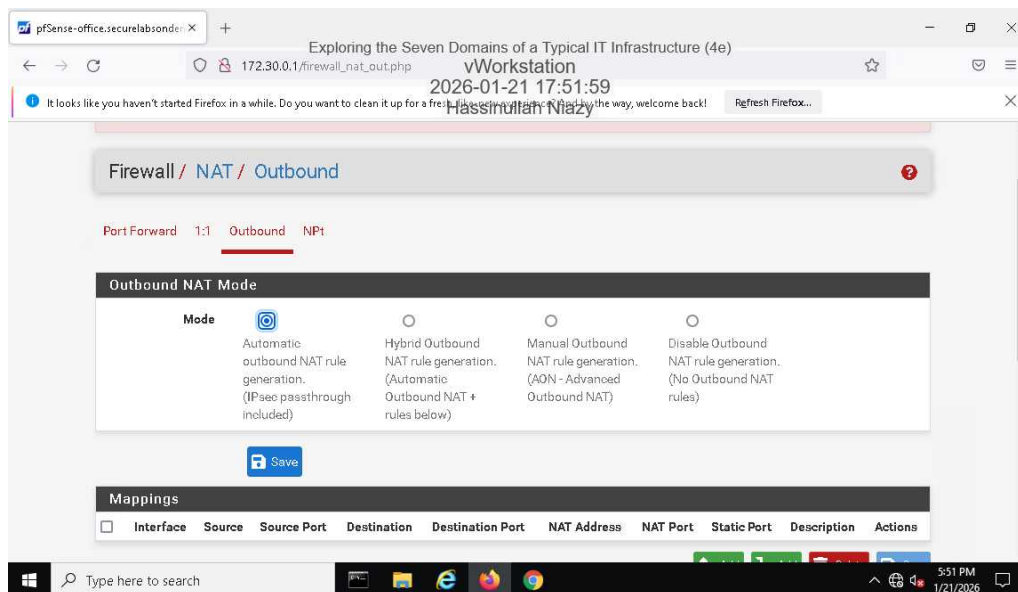


30. Make a screen capture showing the contents of the Employees directory.

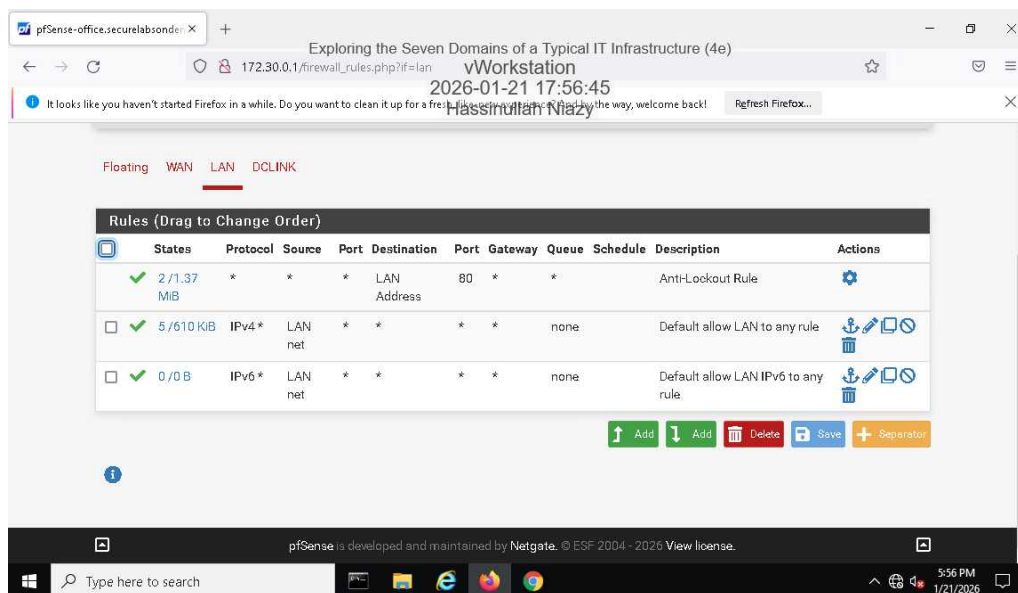


Part 3: Explore the LAN-to-WAN Domain

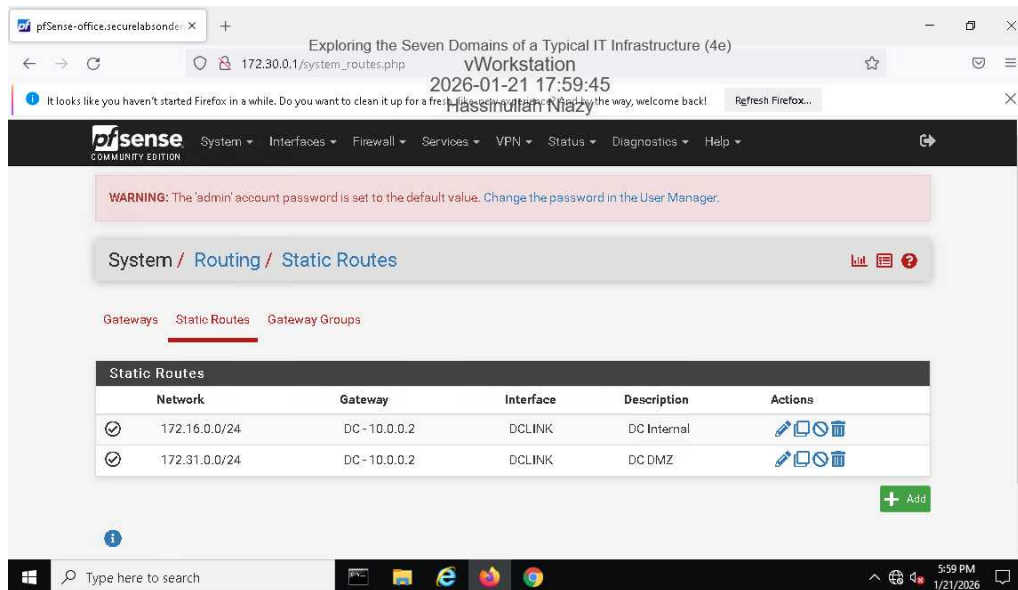
6. Make a screen capture showing the Outbound NAT settings.



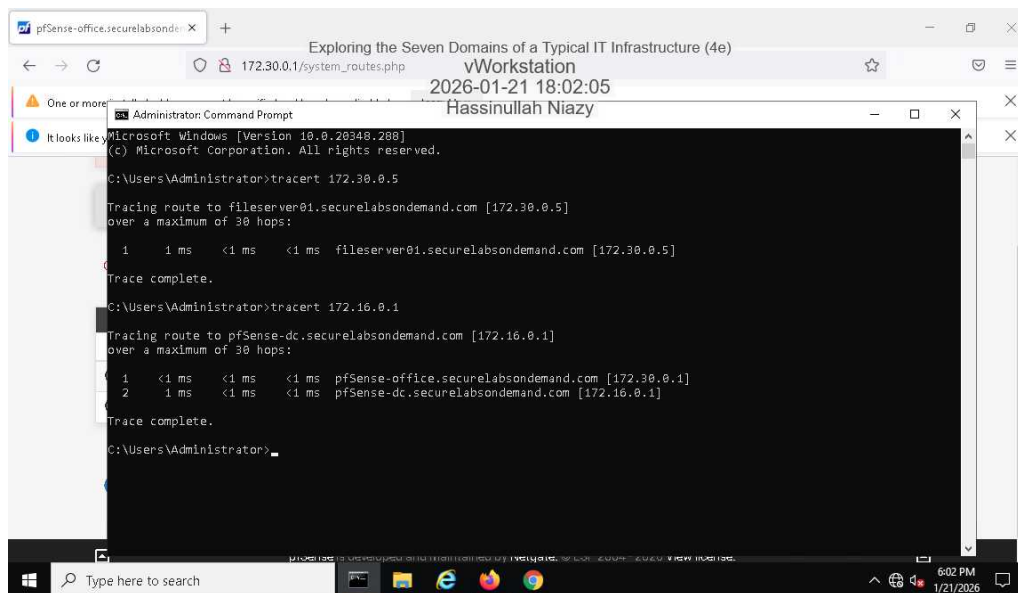
9. Make a screen capture showing the permissive LAN rules.



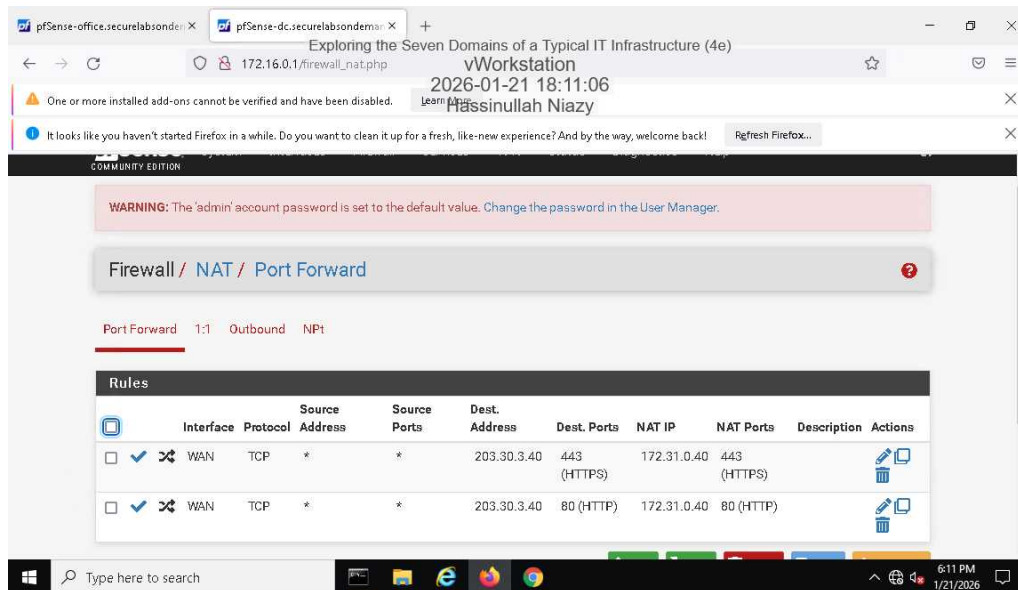
12. Make a screen capture showing the **Static Routes** page.



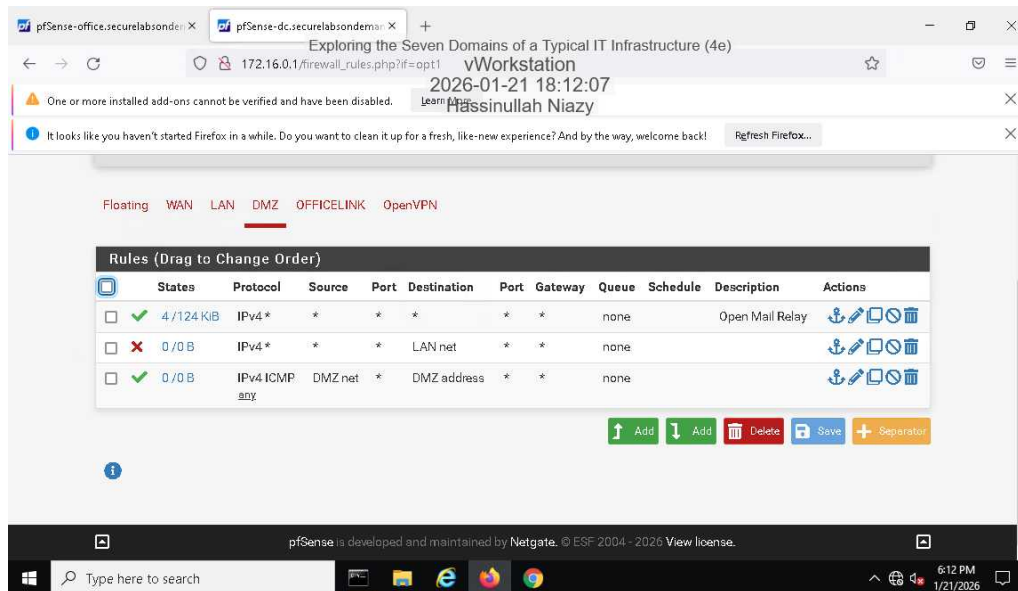
16. Make a screen capture showing the result of your tracer to the pfsense-dc appliance.



22. Make a screen capture showing the Port Forward rules for the web server.



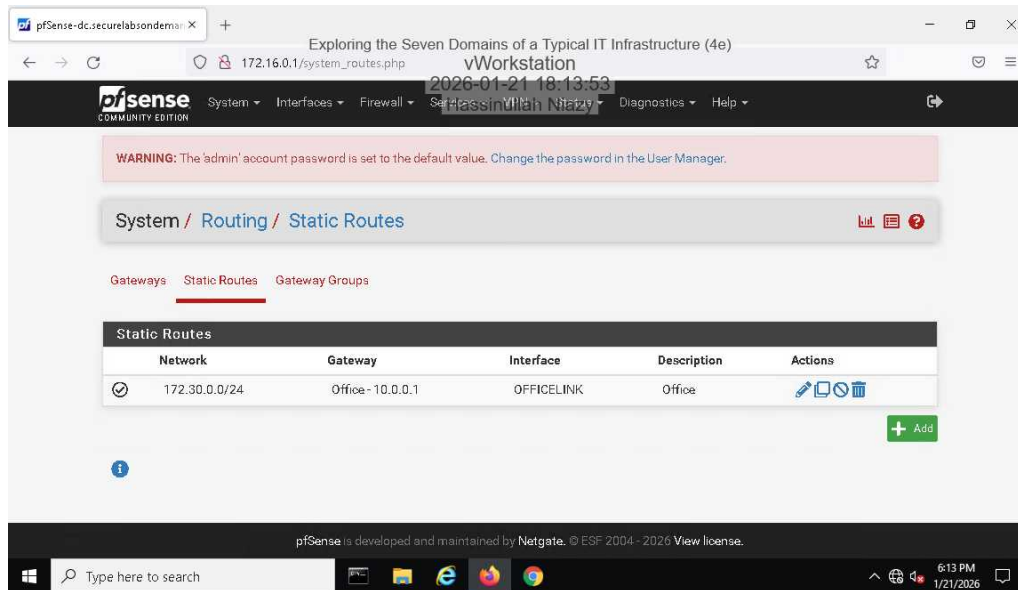
25. Make a screen capture showing the DMZ firewall rules.



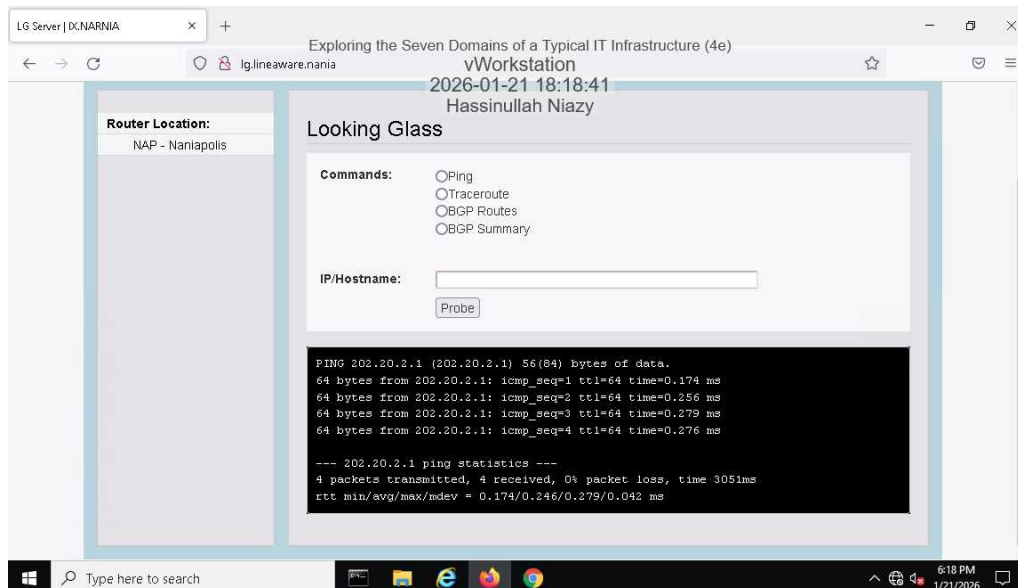
Section 2: Applied Learning

Part 1: Explore the WAN Domain

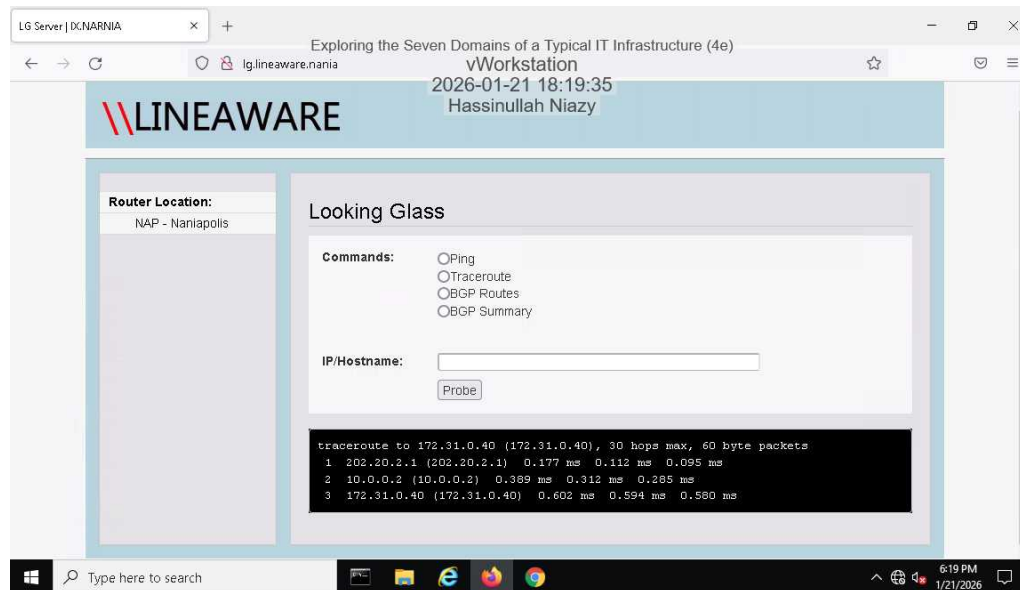
5. Make a screen capture showing the **static route** for the point-to-point connection.



9. Make a screen capture showing the **BPG neighbor ping results**.

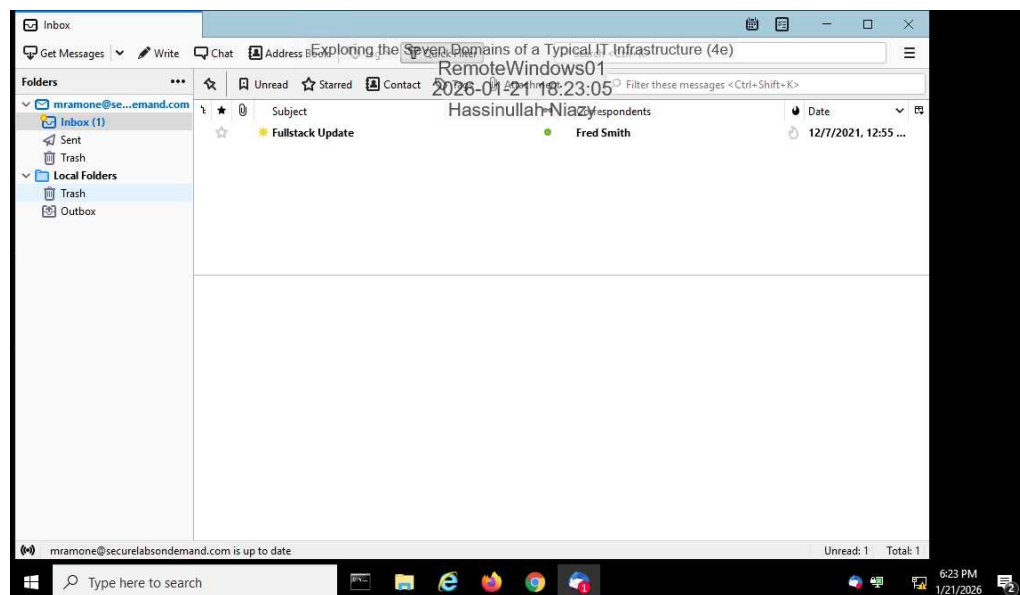


12. Make a screen capture showing the **traceroute** to the file server.



Part 2: Explore the Remote Access Domain

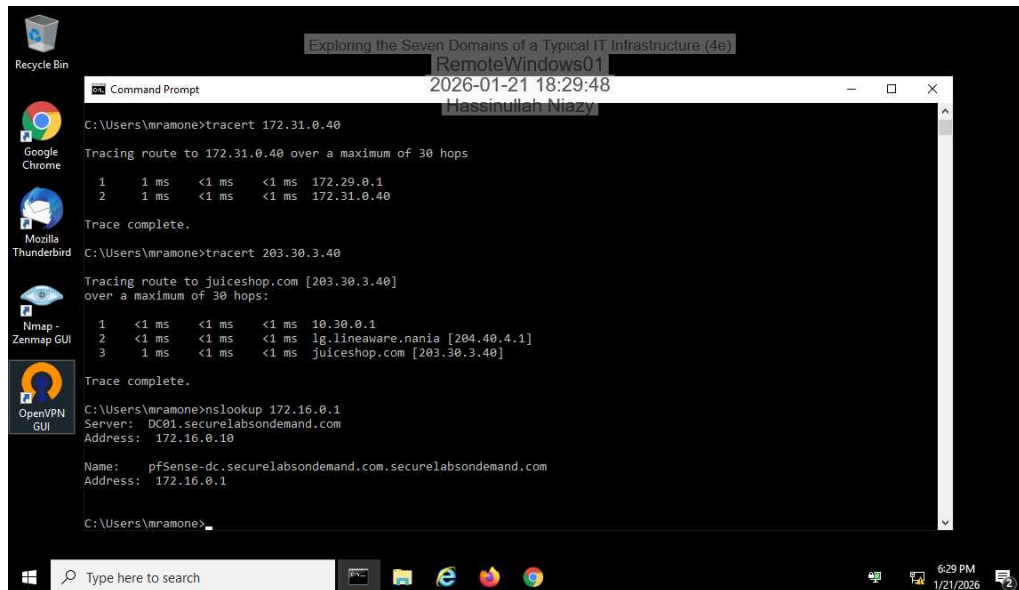
9. Make a screen capture showing the **successful connection** to the email server.



14. **Document** whether the VPN connection is split tunnel or full tunnel, based on the tracert results.

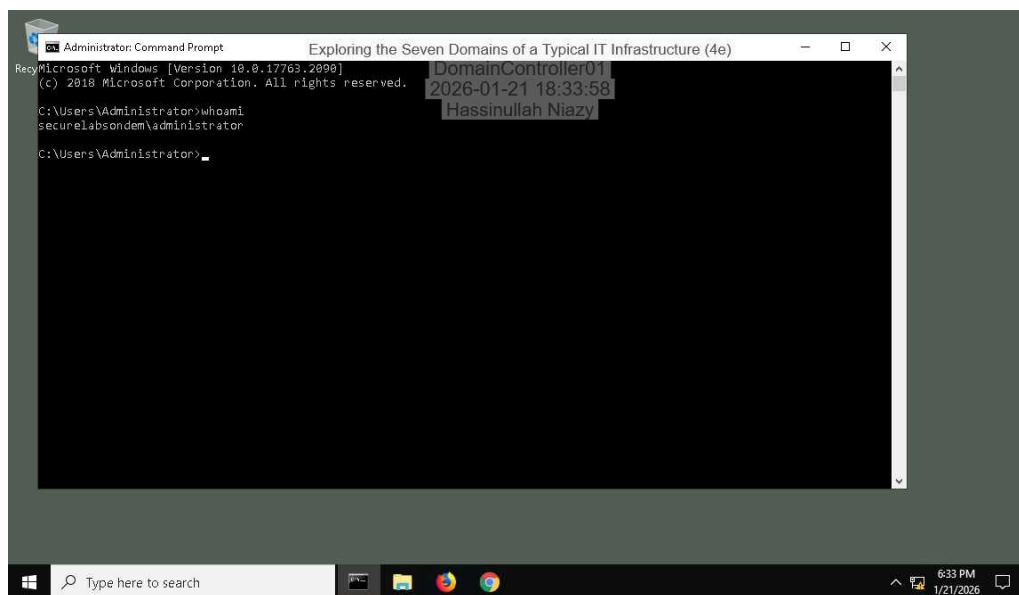
split tunnel

16. Make a screen capture showing the **successful reverse DNS lookup** for the internal host.

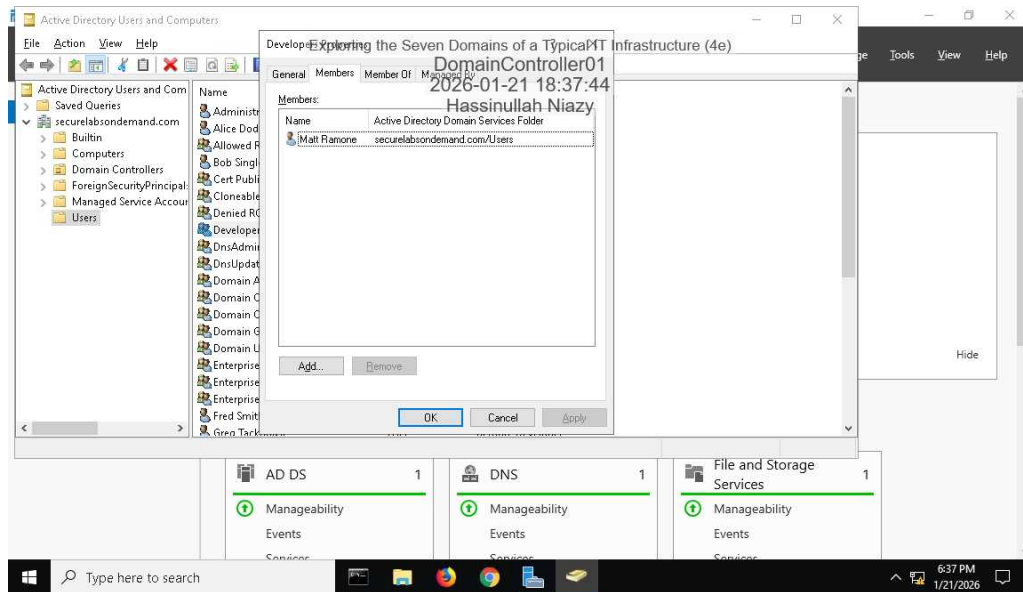


Part 3: Explore the System/Application Domain

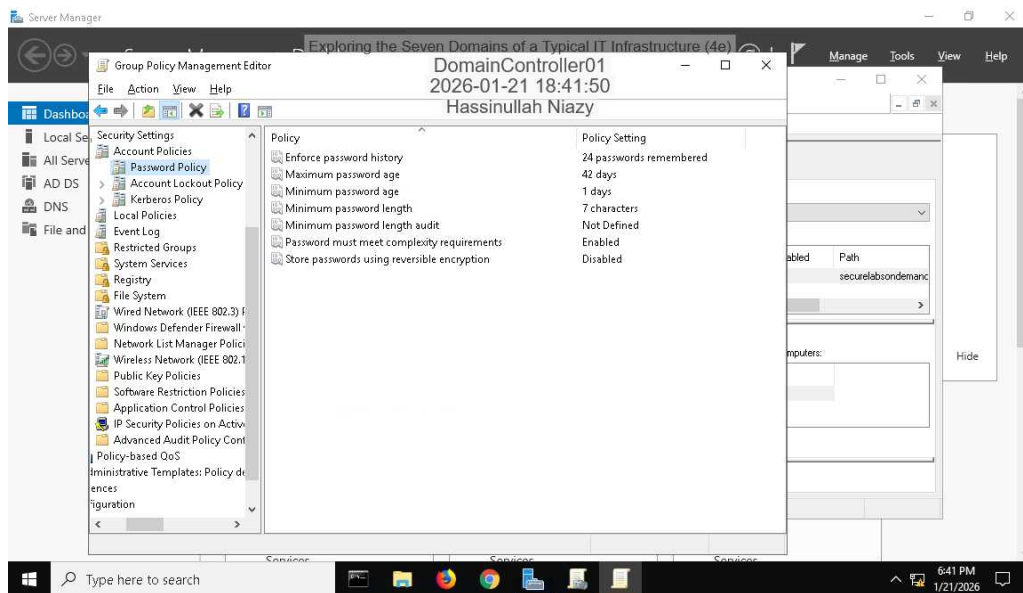
4. Make a screen capture showing the **whoami** results.



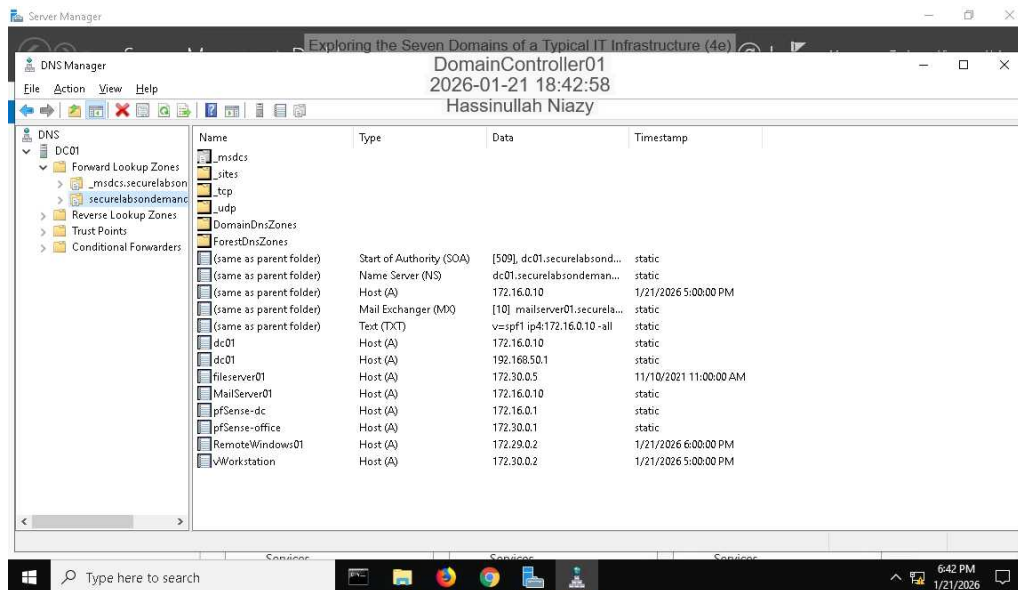
10. Make a screen capture showing the members of the Developers AD group.



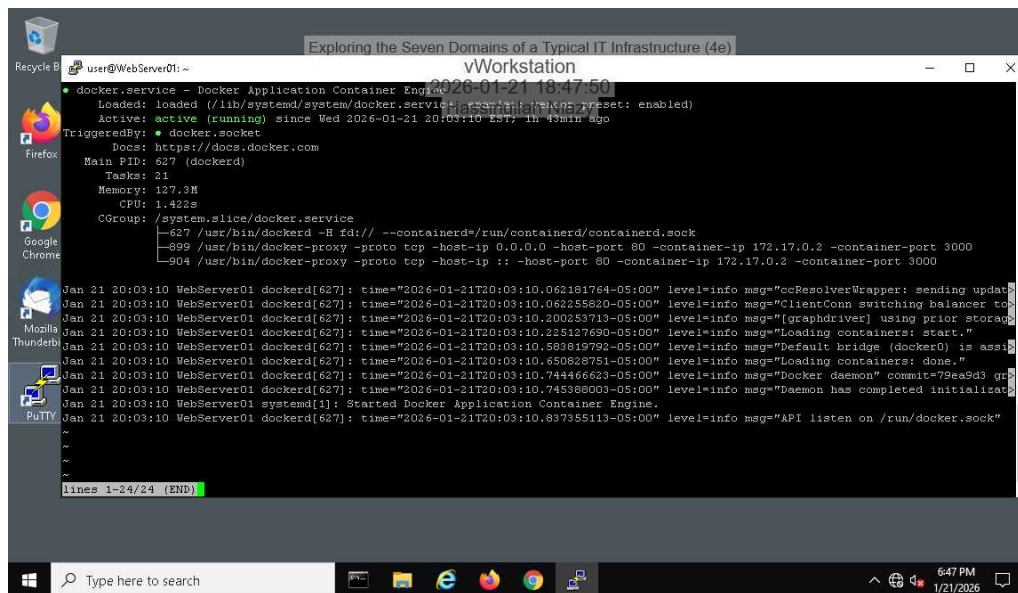
16. Make a screen capture showing the password policy settings in the Group Policy Management Console.



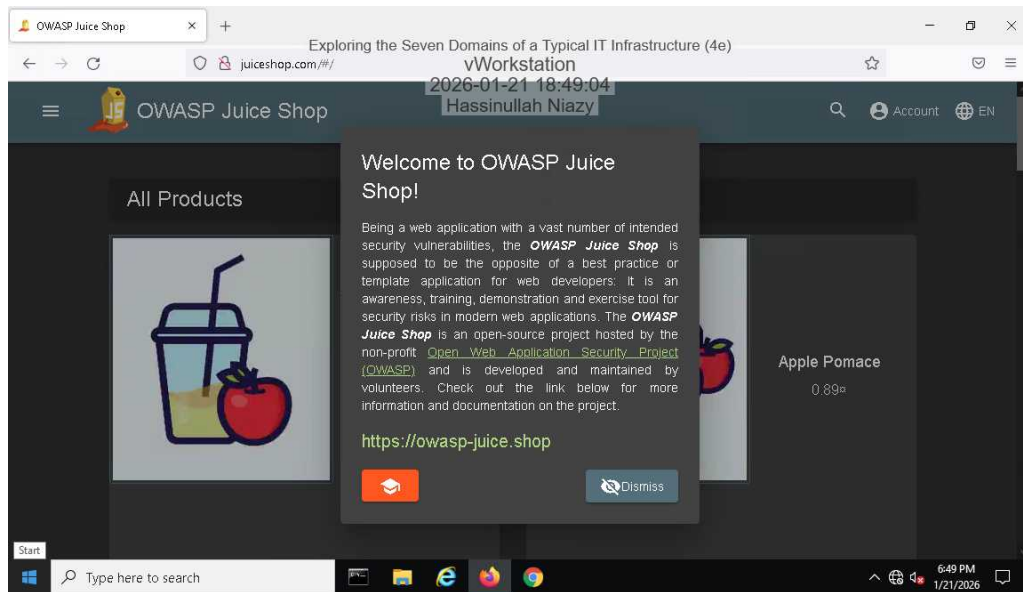
20. Make a screen capture showing the DNS entries.



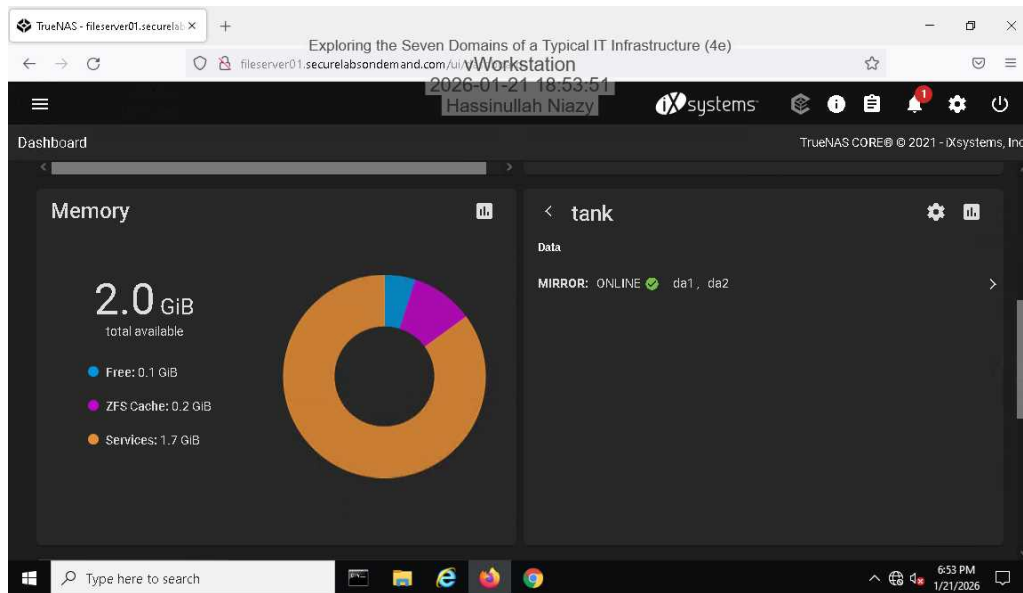
28. Make a screen capture showing the Docker service status.



31. Make a screen capture showing the **juiceshop.com** web page.



36. Make a screen capture showing the disks in the tank volume.



Section 3: Challenge and Analysis

Part 1: Explore the User Domain

Based on your research, **identify** at least **two compelling threats** to the User Domain and **two effective security controls** used to protect it. Be sure to cite your sources.

Social Engineering and Malware are two compelling threats to the User Domain to steal their credential and data. The two effective security controls would be 2-Step Authentication and educating the user about secure browsing and trainings on identifying malicious websites.

Part 2: Research Additional Security Controls

Based on your research, **identify** security controls that could be implemented in the Workstation, LAN, LAN-to-WAN, WAN, Remote Access, and System/Application Domains. **Recommend** and **explain** one security control for each domain. Be sure to cite your sources.

Workstation: Administrative Control LAN: Only Authorized Devices can connect. LAN-to-WAN: Virtual Private Networks (VPNs). WAN: Firewall and SSL. Remote Access: Multi-Factor Authentication and Administrative Control. System/Application Domain: Access Control and Encryption.